Qian Qian

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Education

University of Waterloo Waterloo, Ontario, Canada
Candidate for Bachelor of Software Engineering, Honours Software Engineering, Co-op *September 2025 — May 2030*

Skills

- **Languages**: Java, Kotlin, Python, Rust, C, C++, Shell, Swift, Go, Typst, LaTeX
- **Tech/Tools**: Git, GitHub Actions, Linux, Unit Testing, JUnit, PyTest, NGINX, Docker, Ktor, Flask, SwiftUI, Quilt

Experience

Software Developer & Software Subsystem Co-Lead September 2025 — Present
Waterloo Rocketry *Waterloo, Ontario, Canada*

- Built Mission Control deployment infrastructure to ensure consistent and reproducible software environments using Docker containerization and Ansible configuration management.
- Developed Python-based data acquisition and processing systems to improve reliability of Mission Control telemetry workflows, supported by PyTest unit testing.
- Assembled and tested ground-side electrical and sensor systems to support reliable launch-pad operations during integration and pre-launch testing.
- Mentor developers to improve team-wide engineering practices through development standards, Git workflows, unit testing, pull-request reviews, and collaborative debugging.

Software Engineering Intern, Build and Automation May 2026 — August 2026
Maxon *Montreal, Quebec, Canada*

- Built a Redshift rendering performance-monitoring system to identify regressions across standard rendering unit tests using Python, Grafana, and Loki for data processing, aggregation, and visualization.
- Automated ArchVis changelog generation and build notifications to improve QA visibility using Python and GitLab and Jira APIs.

Projects

QubeC - *Rust, Zig, C3, C++, Go* August 2025 — Present

- Built a bytecode compiler and stack-based virtual machine for a custom C-like programming language.
- Explored multiple programming languages through compiler development and smaller experimental projects to compare language features, tooling, ergonomics, and performance characteristics.

Typstify - *Swift, SwiftUI, Rust* July 2024 — Present

- Developed a native Typst editor for iPadOS to support document authoring and preview workflows using SwiftUI with a Rust-based Typst backend, bridged via swift-bridge for cross-language interoperability.
- Contributed patches to 3 upstream dependencies to expand functionality and resolve existing issues.

Maintainer, ungoogled-chromium-macos - *Shell, Python, Build Systems* November 2023 — Present

- Maintained the macOS distribution of ungoogled-chromium, a privacy-focused browser, delivering 100+ releases to 100,000+ users through upstream tracking, build-script maintenance, and downstream patch management.
- Reduced CI build time by 40% by optimizing GitHub Actions workflows to accelerate build time and release cycles.

CalcLox - *Java* January 2025 — May 2025

- Designed and implemented a Lox-based scripting language and its interpreter for a programmable calculator.
- Built JUnit unit tests to validate language semantics and interpreter correctness across core calculator workflows.

Additional Information

- **Honour and Awards**: Winner of Hack the North 2025 QNX Software Challenge, Canadian Senior Mathematics Contest Distinction (2024), 2 x Canadian Computing Competition Distinction (2023, 2024)
- **Activities**: Programming Competition Problem Setter at CRC Robotics